Dissertation Proposal

Topic: Improving Healthcare Outcomes through Predictive Analytics with Machine Learning

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1. Introduction

The health care system is finding growing problems relating to the precision of forecasting a patient outcome, re-hospitalization and disease. The evidence of the application of the obsolete methods to the prediction of the outcomes depends on the experience of the clinicians so far, and on rule-based methods that, in most situations, would not be able to capture the complexity of the data concerning the patient. It causes sub-efficiencies when planning the treatment and the absence of observing new interventions.

Precise predictive analytics in the health sector can change the patient care process by giving evidence-based advice in a timely manner, which is capable of enhancing the patient care plans and cut down on unwarranted hospitalization. Using machine learning models, large and intricate healthcare data can be processed, and patterns can be identified that otherwise would remain hidden with the help of traditional methods (Badawy et al., 2023). The study is timely in that it might be refined in order to improve decision-making, prescription to tailor made medicine, and lower healthcare spending. In addition, according to the availability of healthcare information further, machine learning may be considered as a feasible and cost-effective approach to the contemporary healthcare problems.

The patients will experience a higher level of certainty when it comes to being diagnosed, less readmission, and treated course-tapping. Physicians will be well informed with the tools to help them in risk stratification that will enhance efficacy and confidence in medical judgment. The planning on a basis of decisions might assist the healthcare facilities in maximum distribution of resources in addition to minimizing costs. The policymakers and other health agencies will also benefit through theoretical guidance on forecasting that will help lead the sculpting of the healthcare policy resulting in finally an enhanced condition of population health and a more sustainable and healthy healthcare system (Dev et al., 2022).

2. Background

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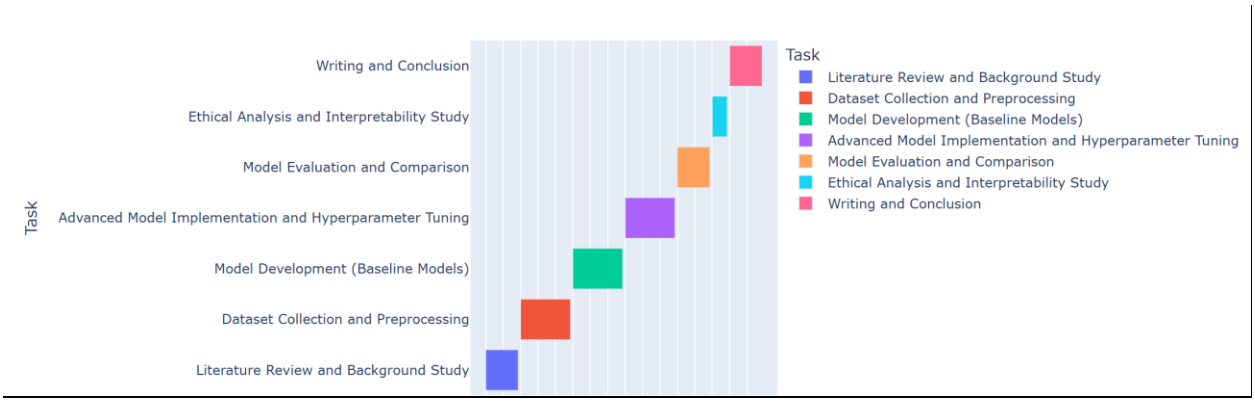
Task	Duration (Weeks)	Timeline
Literature Review & Background Study	2 weeks	Weeks 1–2
Dataset Collection & Preprocessing	3 weeks	Weeks 3–5
Model Development (Baseline Models)	3 weeks	Weeks 6–8
Advanced Model Implementation & Hyperparameter Tuning	3 weeks	Weeks 9–11
Model Evaluation & Comparison	2 weeks	Weeks 12–13
Ethical Analysis & Interpretability Study	1 week	Week 14



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